Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Index **C**
2. Radicand **A**
3. Principal root **F**
4. Conjugate **H**
5. Extraneous solution **E**
6. Composition **B**
7. Inverse function **G**
8. Anchor point **D**

(A) Everything written underneath the radical symbol; when the index is even, it may not be negative.

(B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$.

(C) The small number in the crook of the radical symbol that says which power is being undone.

(D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd.

(E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.

(F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.

(G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$.

(H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. Written with a rational exponent, $\sqrt[3]{x^5}$ is **A**

   (A) $x^{5/3}$ (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$

2. The domain of $f(x) = \sqrt{7 - 2x}$ is **C**

   (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ (C) $x \leq \frac{7}{2}$ (D) all real numbers

3. The graph of $y = \sqrt{3x - 12} + 5$ begins at the point **D** (A) $(12, 5)$ (B) $(4, -5)$ (C) $(-4, 5)$ (D) $(4, 5)$

4. In simplest form, $\sqrt{75} + \sqrt{12}$ is **B** (A) $\sqrt{87}$ (B) $7\sqrt{3}$ (C) $7\sqrt{6}$ (D) $10\sqrt{3}$

5. Which equation has *no real solution*? **Letter: C**

(A) $\sqrt[3]{x+1} = -4$ (B) $\sqrt{x-1} = 7$ (C) $\sqrt{x+6} = -2$ (D) $\sqrt{x} = 0$

6. If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ **D** (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ (D) $\sqrt{x+9}$

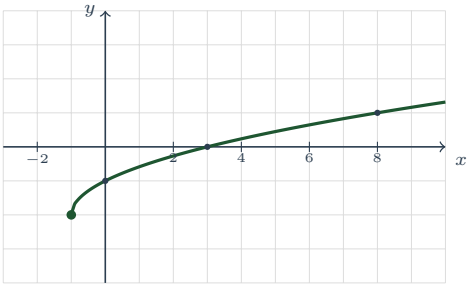
Teacher Note

Item 1: **denominator = index, numerator = power**. (B) is the signature Unit 5 error — the flipped fraction — and (C) multiplies index by power. Item 2: set the *radicand* ≥ 0: $7 - 2x \geq 0$ gives $-2x \geq -7$, and dividing by -2 **flips** the inequality. (A) is the un-flipped answer. Item 3: factor first — $\sqrt{3x - 12} = \sqrt{3}\sqrt{x - 4}$, so the graph starts at $x = 4$, not $x = 12$; (A) is the students who never factored. Item 4: $5\sqrt{3} + 2\sqrt{3}$; (A) adds radicands, (C) multiplies them, (D) multiplies the coefficients. Item 5: an isolated *even* root can never equal a negative number. Distractor (A) is the one to sort for — a student who learned "roots can't be negative" instead of "*even* roots can't" picks it, and that is a whole-unit failure, not a slip; $\sqrt[3]{x+1} = -4$ solves fine ($x = -65$). Item 6: the function next to the x runs first. (A) is the order error, (C) reads $\circ$ as multiplication.

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



(a) Coordinates of the endpoint: (-1,-2)

(b) Domain (interval notation): $[-1, \infty)$; range: $[-2, \infty)$

(c) x-intercept: (3,0); y-intercept: (0,-1)

(d) Interval(s) where f is increasing: $[-1,\infty)$ — the whole graph

(e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

absolute **minimum** -2 at the endpoint $x = -1$; no maximum

(f) End behavior: as $x \to \infty$, $f(x) \to$ ∞; as $x \to -\infty$,

nothing — the graph **stops** at $x = -1$

(g) Interval(s) where $f(x) < 0$: $[-1,3)$

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

(a) Write $\sqrt[4]{x^3}$ with a rational exponent. $x^{3/4}$

(b) Write $y^{5/2}$ as a radical. $\sqrt{y^5}$, or $(\sqrt{y})^5$

(c) $27^{2/3} =$ 9 (d) $16^{-1/2} =$ 1/4

(e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. **Answer:** $\sqrt[6]{x^5}$

Add the exponents: $x^{1/2+1/3} = x^{3/6+2/6} = x^{5/6} = \sqrt[6]{x^5}$. As radicals the two factors could not be combined at all — mismatched indices — which is exactly why the rational-exponent notation exists.

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$ **(a)** $6\sqrt{2}$ **(b)** $3\sqrt[3]{2}$

(a) Largest perfect square: $\sqrt{36 \cdot 2} = 6\sqrt{2}$. (Stopping at $2\sqrt{18}$ is true but unfinished — always check the leftover.)

(b) The index sets the group size, so hunt for a perfect *cube*: $\sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$.

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$ **(c)** $2x\sqrt{5x}$ **(d)** $7\sqrt{3}$

(c) $\sqrt{4x^2 \cdot 5x} = 2x\sqrt{5x}$ — divide the exponent by the index: x^3 gives one x out and one x left in.

(d) Simplify first, decide second: $5 \cdot 2\sqrt{3} - 3\sqrt{3} = 10\sqrt{3} - 3\sqrt{3} = 7\sqrt{3}$. Before simplifying these look like unlike radicals.

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6} \cdot \sqrt{15}$ (b) $(2 + \sqrt{5})(2 - \sqrt{5})$ **(a)** $3\sqrt{10}$ **(b)** -1

(a) $\sqrt{90} = \sqrt{9 \cdot 10} = 3\sqrt{10}$. Neither factor simplifies on its own — the *product* does, so the last step is the graded one.

(b) Conjugates: $4 - 5 = -1$, a rational number. Compare $\left(2 + \sqrt{5}\right)^2 = 9 + 4\sqrt{5}$, where the radical survives.

(c) $\dfrac{6}{\sqrt{3}}$ (d) $\dfrac{4}{3 - \sqrt{5}}$ **(c)** $2\sqrt{3}$ **(d)** $3 + \sqrt{5}$

(c) Multiply by $\frac{\sqrt{3}}{\sqrt{3}} = 1$: $\dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$.

(d) Multiply by the *conjugate* $\frac{3+\sqrt{5}}{3+\sqrt{5}}$: $\dfrac{4\left(3 + \sqrt{5}\right)}{9 - 5} = \dfrac{4\left(3 + \sqrt{5}\right)}{4} = 3 + \sqrt{5}$. Using $3 - \sqrt{5}$ again would only produce a new radical.

5. **Transformations.** Let $g(x) = -2\sqrt{x + 3} + 4$.

(a) List every transformation that takes $y = \sqrt{x}$ to g.

left 3, stretch $\times 2$, reflect over x-axis, up 4

(b) Anchor point: (-3,4)

(c) Domain: $[-3, \infty)$; range: $(-\infty, 4]$

(d) Is the anchor an absolute maximum or an absolute minimum? maximum ($a < 0$ turns the arm over)

(e) A different graph of the form $y = \sqrt{x - h} + k$ begins at $(5, -2)$. Write its equation. $y = \sqrt{x-5} - 2$

(f) For $p(x) = \sqrt[3]{x - 1} + 2$, give the anchor point (1,2) and the domain $(-\infty, \infty)$. In one sentence, say why that domain is not like g's.

index 3 is odd: cubing keeps the sign, so no input is forbidden

6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x + 3} = x$ **Answer:** $x = 3$ only

Square: $2x + 3 = x^2$, so $x^2 - 2x - 3 = 0$ and $(x - 3)(x + 1) = 0$, giving candidates $x = 3$ and $x = -1$. Check $x = 3$: $\sqrt{9} = 3$ ✓. Check $x = -1$: $\sqrt{1} = 1$, but the right side is -1, so $x = -1$ is extraneous. Solution: $x = 3$ only. (Note the tell — the non-radical side is negative at the rejected value.)

(b) $\sqrt[3]{x - 4} = -2$ **Answer:** $x = -4$

Cube both sides — legal in both directions, since every real number has exactly one real cube root: $x - 4 = -8$, so $x = -4$. Check: $\sqrt[3]{-8} = -2$ ✓. No extraneous risk with an odd index.

(c) $\sqrt{3x-2}+4=1$ **Answer:** no real solution

Isolate first: $\sqrt{3x-2}=-3$. An isolated *even* root is never negative, so no real solution — and no algebra was needed. (Squaring anyway gives $x=\frac{11}{3}$, a phantom that fails the check.)

(d) $x^{2/3}=25$ **Answer:** $x=\pm 125$

Raise both sides to the reciprocal power $\frac{3}{2}$: $x=\pm 25^{3/2}=\pm 125$. The *numerator* 2 is even — it is a square in disguise — so write the $\pm$ yourself. Both check: $(\pm 125)^{2/3}=(\pm 5)^2=25$ ✓

7. **Composition.** Let $f(x)=\sqrt{x}$ and $g(x)=x-6$.

(a) $(f\circ g)(x)=$ $\sqrt{x-6}$, domain: $x\geq 6$

(b) $(g\circ f)(x)=$ $\sqrt{x}-6$, domain: $x\geq 0$

(c) Are these the same function? Answer using *both* the rule and the domain.

No: different rules (right 6 vs down 6) and different domains ($x\geq 6$ vs $x\geq 0$)

(d) $(f\circ g)(10)=$ $\sqrt{4}=2$

(e) For $p(x)=2x+1$ and $q(x)=x^2-3$, find $(p\circ q)(-2)$. $q(-2)=1$, then $p(1)=3$

8. **Inverses.**

(a) $f(x)=4x-7$. Find $f^{-1}(x)$. $f^{-1}(x)=(x+7)/4$

(b) Verify by composition — *both* orders — that your answer really is the inverse.

$f(f^{-1}(x))=x$ and $f^{-1}(f(x))=x$: both orders return x

(c) $g(x)=\sqrt{x+2}$. Find $g^{-1}(x)$ and state its domain. $g^{-1}(x)=$ x^2-2 **domain:** $x\geq 0$

Swap and solve: $x=\sqrt{y+2}$, so $x^2=y+2$ and $y=x^2-2$. Domain: $x\geq 0$, because that is the *range* of g — read it off, never guess it.

(d) The point $(7,3)$ lies on the graph of g. What point must lie on the graph of g^{-1}? (3,7)

Teacher Note

Item 1(f) is the Unit 5 habit-breaker: "the graph stops" is a *complete* answer to the left-end question, not a missing one — do not accept "$-\infty$" or a blank. Item 1(g): the endpoint is included, so $[-1,3)$; a student writing $(-1,3)$ has forgotten that $\sqrt{0}$ is real. Item 2(e): $x^{5/6}$ alone earns most of the credit, but the item asked for a radical. Item 3: an answer that is simplified but not *fully* simplified (e.g. $2\sqrt{18}$) is half an answer. Item 4(d): watch for the a^2-b^2 sign flip in the denominator. Item 5(d) is the most-missed — a negative a makes the

anchor a **maximum**, and the range flips with it; 5(f) must name the *index*, not just assert "all reals." Item 6(a) is the graded check: "$x = 3$ and $x = -1$" as a final answer earns no credit, because the check is the step being assessed — and 6(c) should be answered *before* any squaring. Item 7(c) must cite the domain, not only the rule; "they look different" is not the argument. Item 8(b) is A2.F.2i's justification clause — both orders, or it is not a verification.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x-intercept and the y-intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f. **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f.

(a) Anchor $(1, 2)$; domain $[1, \infty)$; range $[2, \infty)$. Both start at the anchor.

(b) **Neither exists.** An x-intercept would need $\sqrt{x-1} + 2 = 0$, i.e. $\sqrt{x-1} = -2$, and an even root is never negative. A y-intercept would need $f(0)$, but 0 is not in the domain — the graph does not reach the y-axis.

(c) Right 1, up 2. ($a = 1$: no stretch, no reflection.)

(d) Swap and solve: $x = \sqrt{y-1} + 2$, so $x - 2 = \sqrt{y-1}$, $(x-2)^2 = y - 1$, and $f^{-1}(x) = (x-2)^2 + 1$ for $x \geq 2$. The restriction is the **range of** f, written down: whatever f can output is exactly what f^{-1} can accept.

(e) $f\big(f^{-1}(x)\big) = \sqrt{(x-2)^2 + 1 - 1} + 2 = \sqrt{(x-2)^2} + 2 = |x-2| + 2$, and since $x \geq 2$ that is $(x-2) + 2 = x$. Going the other way, $f^{-1}(f(x)) = \big(\sqrt{x-1} + 2 - 2\big)^2 + 1 = \big(\sqrt{x-1}\big)^2 + 1 = (x-1) + 1 = x$ for $x \geq 1$. Both orders return x, so the two functions undo each other.

(f) Without the restriction it is the *whole* parabola — two arms — and it fails the horizontal line test, so it cannot be the inverse of anything: it sends both $x = 0$ and $x = 4$ to 5. It also accepts inputs f never produces; f never outputs a number below 2, so $x = 0$ has no business being fed to f^{-1}. Only the right arm, $x \geq 2$, is the inverse.

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

(a) State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse, h as a function of v, and say what it tells an engineer that $v(h)$ does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.

(a) $h \geq 0$, i.e. $[0, \infty)$. The index is even, so the radicand may not be negative — and the context says the same thing twice over, since a drop of negative height is not a thing. Here the algebra and the situation agree; that is not always true.

(b) $v(9) = 8\sqrt{9} = 8 \cdot 3 = 24$ feet per second; $v(36) = 8\sqrt{36} = 8 \cdot 6 = 48$ feet per second. Quadrupling the drop only **doubles** the speed, because $\sqrt{4h} = 2\sqrt{h}$ — an exponent of $\frac{1}{2}$ flattens growth, which is the shape of the square root graph.

(c) $8\sqrt{h} = 56$, so $\sqrt{h} = 7$ and $h = 49$ feet. Check: $v(49) = 8\sqrt{49} = 8 \cdot 7 = 56$ ✓

(d) Solve for h: $\sqrt{h} = \frac{v}{8}$, so $h = \dfrac{v^2}{64}$ for $v \geq 0$. The original answers "how fast will this drop make the car go?"; the inverse answers the question an engineer actually asks — "how tall must I build the drop to hit a target speed?"

(e) An extraneous root only appears when squaring hides a *sign* disagreement — the step $a^2 = b^2$ forgets whether $a = b$ or $a = -b$. Here both sides are non-negative: $\sqrt{h} \geq 0$ and a speed of 56 is positive, so there is no hidden sign to lose and the single candidate must be genuine. Checking it is still good practice.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) anchor with *both* interval-notation answers; 1 pt (b) *both* non-existences with reasons — "no x-intercept" with no reason earns half; 1 pt (c) both transformations; 1 pt (d) the rule *and* $x \geq 2$ *and* the sentence that it is the range of f (no credit for the restriction if it is guessed as $x \geq 0$, the reflex answer); 1 pt (e) both compositions — one order is not a verification, and this is the standard's own "justify" clause; 1 pt (f) the horizontal line test or the two-arms argument. A student who writes $(x-2)^2 + 1$ with no restriction in (d) and then cannot explain (f) has missed the unit's closing idea — score both accordingly. D2: 1 pt (a) $h \geq 0$ tied to the even index; 1 pt (b) both values with units *and* "doubles" with $\sqrt{4h} = 2\sqrt{h}$ (half credit for "it doubles" with no reason); 1 pt (c) $h = 49$ ft; 1 pt the check; 1 pt (d) $h = \frac{v^2}{64}$; 1 pt (e) the sign argument. Do not accept "because I checked it" for (e) — the question asks why the danger was never there.